

Stride — System Design Blueprint

From the final proposal to a production-grade system *Performance Prediction and Race-Condition Nutrition Estimation for Professional Long-Distance Runners*

- **Status:** design blueprint v1.0 — architecture only, nothing implemented
- **Source of truth:** the final project proposal (Sections 1–6)
- **Working name:** "Stride" is OUR APP NAME

How to use this blueprint

1. Read **S 1** (what the proposal commits to) and **S 14.3** (open decisions) first. They set the limits every other section respects.
2. Sections **1–14 follow the requested outline one-to-one**. **S 15** turns the design into a step-by-step build path.
3. Every capability carries an **evidence tag**, so the product never promises more than the proposal has established.

Tag	Meaning	Example
[V]	Validated pilot: run on real data, metrics reported in the proposal	Marathon finish-time MLP
[P]	Planned by the proposal, not yet validated	CBR adaptations, nutrition engine
[C]	Context or descriptive analytics, no model involved	Weekly volume, resting-HR trend
[X]	Extension beyond the proposal, optional and clearly separated	Weather panel, device connectors

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1. System overview

1.1 Purpose

A web platform where professional long-distance runners and their coaches turn training and physiology data into three things: a **projected marathon finish time**, **training adaptations drawn from similar runners' histories**, and a **periodized nutrition and race-fueling plan** whose targets

follow the prescribed training load and never exceed what the gut can absorb (≤ 90 g/h). This is the proposal's coupled dual task: Task 1A (performance and training) feeds Task 1B (nutrition).

1.2 What the proposal commits to

Proposal element	System capability	Tag
MLP: split_min, age, gender → marathon finish time	Race Predictor (from a 21.1 km marathon split)	[V] MAE 14.21 min, RMSE 18.59, R ² 0.916 on the NYC hold-out (single 80/20 split, not cross-validated)
CBR / k-NN retrieval over weekly training vectors; goal delta $\pm 5\%$; volume change $\leq 10\%$	"Runners like you" and training adaptations	[P]
Frayn oxidation, Margaria/Minetti energy cost	Fuel Lab calculations	[P]
Periodized daily macronutrients, pre / in / post competition	Fuel Lab phase view	[P]
SGLT1 and GLUT5 absorption ceiling ≤ 90 g/h	Hard cap on in-race intake	[P]
RQ4: glycogen deficit, time-to-fatigue (TTF) > 120 min	Glycogen budget and flags	[P]
PMData, Fitbit: recovery, resting physiology, NEAT	Descriptive physiology trends	[C]
Strava-origin dumps: grade-adjusted pace (GAP), TRIMP	Per-activity metrics	[C], provenance check needed (S6.7)
coupled_system_eval.py	Plan orchestrator (its production twin)	[P]
Streamlit prototype, coach and athlete views	Internal research dashboard; informs the production UI	—

1.3 Users and roles

Role	Job to be done	Notes
Athlete (primary)	See a projected finish, understand how to adapt training, get a fuel plan for a specific race	Professional or career-focused distance runner

Role	Job to be done	Notes
Coach (co-primary)	Monitor athletes, review and approve recommendations, adjust plans	Sees an athlete's data only after that athlete consents
Admin / ML-ops (internal)	Publish model bundles, manage dataset licences, feature flags, audit	Separate, role-gated console
Visitor	Understand the product; try the public Race Predictor	Unauthenticated, rate-limited
<i>Official / organiser</i>	Not designed	The proposal defines no output for this role

1.4 Scope boundaries

In scope (v1): account and consent management; activity logging and analytics; finish-time projection **for the marathon only**; CBR adaptations; nutrition plans for 5K to marathon distances; race plans; coach review workflow; exports.

Deliberately out of scope:

- Finish-time prediction for 5K, 10K or half marathon (the validated model covers the marathon only).
- Weight-loss features or calorie-restriction advice.
- Generative AI in the decision path (templates only, see S9.6).
- Medical claims or injury prediction.
- Public social feeds (privacy and scope).
- Users under 18 (the models were built on adult data).

Extension slots, designed for but not built: race-day weather as a model input, footwear feature, course-specific modeling, device connectors, feedback-driven retraining (S14.4).

1.5 Design principles

1. **One science, two consumers.** The same code computes results in research notebooks and in production, checked by golden tests.
2. **Evidence before polish.** Every output shows its evidence tag and model version.
3. **Decision support, not decisions.** Coaches approve; athletes can override; hard bounds are enforced in code.
4. **Explain with cases, not jargon.** Show similar runners and sensitivity, not opaque scores.
5. **Race-day robustness.** The plan works offline and on paper.
6. **Privacy by default.** Consent, minimization, export and delete are core features.
7. **Train is not serve.** Models ship as immutable, versioned, verified bundles.

8. Quiet interface, one memorable instrument (the Race Ribbon, S4).

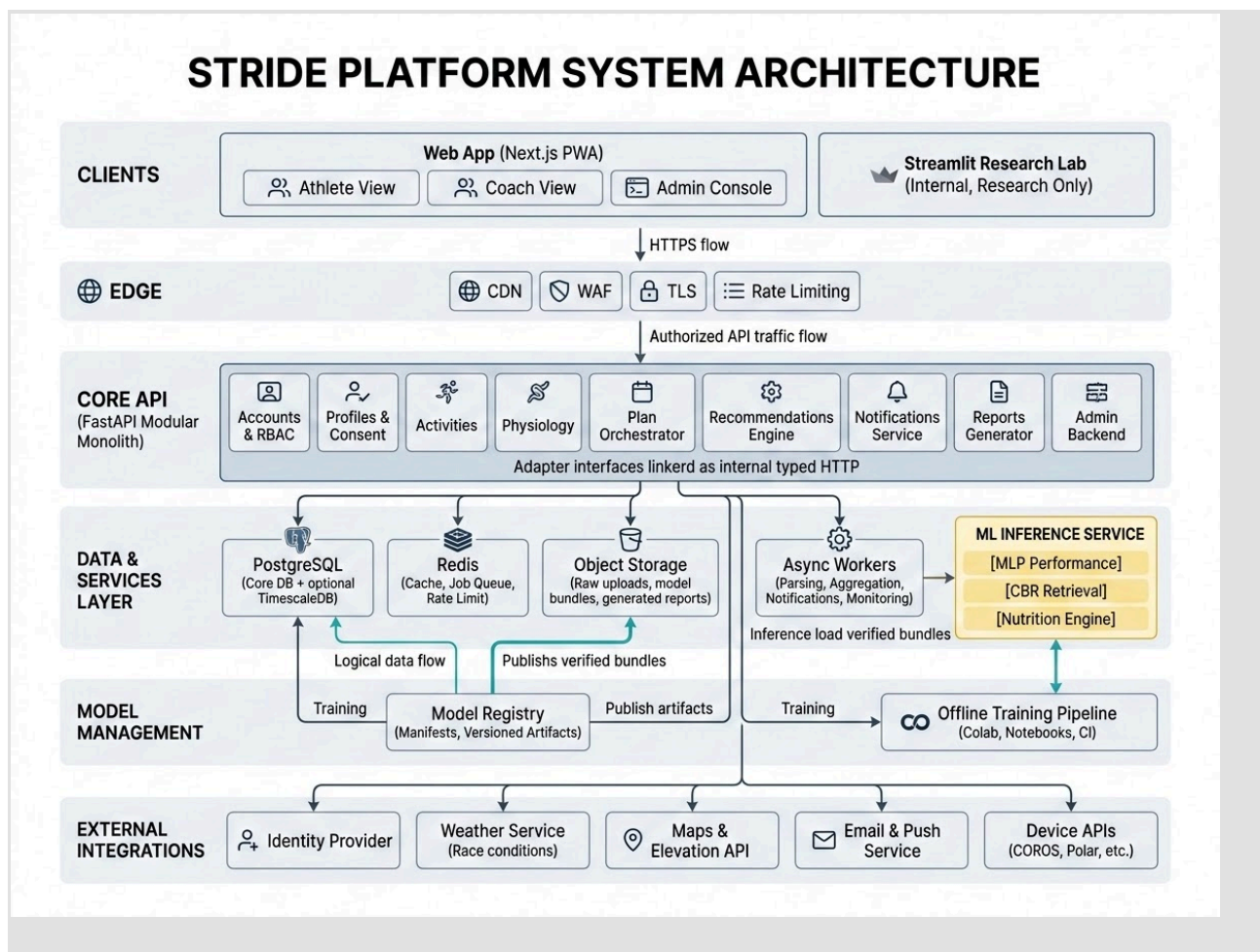
1.6 Grounding checks that shape the design

These were verified while preparing this blueprint (September 2026). They constrain what the product may claim.

- **Afonseca et al. training dataset** documents these columns: datetime, athlete, distance, duration, gender, age_group, country. There is **no finish-time column**, and the records were obtained by web-scraping an athlete social network. Consequence: the CBR engine can say what similar runners *did next*, not what improved their time, until an outcome-labelled dataset exists (Decision D1).
- **PMDData** is licensed **CC BY-NC 4.0** (the licence file bundled with the dataset), has 16 participants over 5 months, and its file listing contains no HRV file. Consequence: non-commercial use only; HRV cannot be assumed from the training corpora.
- **Strava's API policy (effective 1 June 2026)** prohibits using its data for AI development, training, evaluation or operation. Consequence: Strava is not a data source or connector.
- **The proposal's own Table 4:** only the MLP row is a measured result. The CBR row is an unmeasured target, so it is tagged **[P]** here.

2. Overall architecture

2.1 Architecture at a glance



2.2 Components and responsibilities

Component	Responsibility	Scales by
Web app	UI, PWA offline mode, client-side validation, charts	CDN, static and edge rendering
Core API	Business logic, auth enforcement, orchestration, recommendation rules	Horizontal replicas (stateless)
ML inference service	Runs the three engines against verified model bundles; holds no user identity	Horizontal replicas; CPU only
Workers	File parsing, aggregation, notifications, reports, monitoring	Queue depth
PostgreSQL	System of record, row-level security	Vertical, then read replicas
Redis	Cache, job queue, rate limits	Managed service
Object storage	Raw uploads, generated reports, model bundles	Effectively unbounded
Model registry	Versioned bundles plus manifests (model cards)	—
Offline training	The proposal's scripts, run in Colab, notebooks or CI	Not part of the runtime

2.3 How the parts talk

From → To	Protocol	Contract	Notes
Browser → API	HTTPS, REST/JSON; SSE for job progress	OpenAPI 3.1, generated TypeScript client	HttpOnly SameSite cookies or short-lived tokens
API → Inference	Internal HTTP/JSON	Versioned routes, typed schemas	Timeouts, safe retries, circuit breaker
API → Postgres	SQL	Migrations under version control	Row-level security enforced
API → Workers	Redis queue	Job schemas with idempotency keys	Retries with back-off
Inference → Registry	Read-only object storage	Bundle manifest with checksum	Verified before load
API ↔ Externals	HTTPS	Adapter interfaces (ports)	Provider can be swapped

2.4 Key architecture decisions

Decision	Choice	Why	Revisit when
API shape	Modular monolith	Small team, one deployable, clear module boundaries	A module needs independent scaling or release
ML serving	Separate stateless service	Isolates pinned scikit-learn, private network, independent releases	—
Domain science	One shared Python package for notebooks and service	Prevents research/production drift	—
Persistence	PostgreSQL first; files in object storage	Relational integrity and row-level security; cheap durable files	Time-series volume demands a specialised store
Frontend	Next.js PWA	Server rendering for the landing page; offline race-day plan	—
Realtime	Server-sent events for job status	Simple, proxy-friendly	Live race tracking is added
Explanations	Similar cases plus sensitivity; no generative AI	Matches the proposal's XAI; avoids unsupported outputs	Outputs are validated and a narration layer is justified
Model format	scikit-learn Pipeline in a signed bundle; ONNX optional	Parity with research code	Python-version coupling becomes painful

3. Frontend features and pages

3.1 Sitemap

Section	Path / Route	Description / Features
Public	/	Landing page with the live Race Predictor
	/how-it-works	Three-step explainer
	/science	Methodology, model cards, evidence tags, references
	/data	Datasets, provenance, licences, limitations
	/privacy, /terms, /contact, /login, /signup, /reset-password, /verify-email	Authentication and legal pages

Section	Path / Route	Description / Features
Athlete App (/app)	/today	Overview and next action
	/activities, /activities/new, /activities/:id	Activity feed, upload/manual entry, and detailed view
	/train	Projection, runners like you, what-if analysis
	/fuel	Fuel Lab (pre / in race / post)
	/plans, /plans/:id	Race plans and plan detail (Race Ribbon)
	/body	Resting HR, sleep, calories, HRV where provided
	/results	Race results and prediction calibration
	/reports	Weekly and monthly summaries
	/notifications	Alerts and notification center
	/settings	Profile, units, privacy, connections, notifications

3.2 Feature catalogue

Backend codes are defined in S5.1. Release: **MVP** = first usable product, **v1** = full proposal scope, **v1.5** = after v1.

ID	Feature	What it does	Tag	Release	Backend
F1	Sign-up and login	Email verification, optional MFA, athlete or coach role	—	MVP	ACC
F2	Onboarding wizard	Profile (age, sex category, body mass), units, consent, first data source	—	MVP	PRF, CON
F3	Consent center and	Granular consents, data	—	MVP	CON, AUD

ID	Feature	What it does	Tag	Release	Backend
	data rights	export, account deletion			
F4	Coach sharing	Athlete invites a coach, chooses scopes, revokes any time	—	v1	COA
F5	File upload	FIT, GPX, TCX, CSV with progress and a validation report	[C]	MVP	ACT
F6	Quick log	Manual entry of a run	[C]	MVP	ACT
F7	Activity feed and calendar	Filters by type, date, tag; weekly ledger	[C]	MVP	ACT, AGG
F8	Activity detail	Map, elevation, pace/HR, splits, GAP, TRIMP	[C]	MVP (map in v1)	ACT
F9	Weekly aggregates	Distance, duration, run count, long run (the CBR's inputs)	[C]	MVP	AGG
F10	Privacy zones and GPS trimming	Hide start and end areas of a route	—	v1	ACT, PRF
F11	Duplicate and timezone handling	One run from two sources merges; local time stays correct	—	MVP	ACT
F12	Device connectors	COROS or Polar sync, only after a terms check (S7)	[X]	v1.5	INT

ID	Feature	What it does	Tag	Release	Backend
F13	Physiology trends	Resting HR, sleep, calories against a personal baseline band	[C]	v1	PHY, AGG
F14	HRV entry and import	Only where the source provides it	[C]	v1	PHY
F15	Race Predictor	21.1 km marathon split + age + category → projected finish, typical error, sensitivity	[V]	MVP	PRD
F16	Prediction history and calibration	Predicted against actual on the athlete's own results	[V]/[C]	v1	PRD, AGG
F17	Runners like you	Anonymized similar cases from the case base	[P]	v1	CBR
F18	What-if explorer	Adjust volume within the $\pm 5\%$ goal delta and see candidate adaptations ($\leq 10\%$ cap)	[P]	v1	CBR, REC
F19	Load trend and change flag	Week-over-week volume against the proposal's 10% bound	[C]	v1	AGG, REC
F20	Fuel Lab: daily targets	Periodized carbohydrate (and other engine-defined macros) by phase	[P]	v1	NUT
F21	In-race planner	Intake schedule with	[P]	v1	NUT

ID	Feature	What it does	Tag	Release	Backend
		a hard 90 g/h cap and a glycogen trace			
F22	Course energy cost	Distance and average grade → energy cost	[P]	v1	CRS, NUT
F23	Fuel card export	Print, PDF, wristband layout and an offline PWA copy	[P]	v1	RPT
F24	Scenario compare	Plan A against plan B, side by side	[P]	v1.5	NUT
F25	Post-run fueling log	Intake and gut comfort, for review and future calibration	[X]	v1.5	PHY
F26	Race Plan	Race, goal, projection, adaptations, nutrition and timeline in one plan	[P]	v1	PLN, REC
F27	Plan versions	Change log and version compare	—	v1	PLN
F28	Coach roster	Athletes, data freshness, alerts	—	v1	COA, AGG
F29	Review queue	Coach approves, edits or rejects drafts; athlete is notified	—	v1	REC, COA, NTF
F30	Group analytics	Team volume and adherence	[C]	v1.5	AGG
F31	Notes and audit trail	Comments on plans; who	—	v1	AUD

ID	Feature	What it does	Tag	Release	Backend
		changed what and when			
F32	Evidence badges and model cards	Tag, version, metrics and limits on every output	—	MVP	ADM, PRD
F33	Notifications	In-app, email, push; quiet hours	—	v1	NTF
F34	Reports	Weekly summary, race-plan PDF	—	v1	RPT
F35	Weather context	Forecast for race place and date, display only, never a model input in v1	[C][X]	v1.5	ENV
F36	Help and glossary	Plain-language explainers (SGLT1, RER, TTF)	—	v1	—
F37	Admin console	Model registry, dataset and licence register, flags, audit	—	MVP	ADM, AUD

3.3 Landing page blueprint

Goal: a visitor understands the product in ten seconds and can try the real model without signing up.

Section	Content	Notes
Header	Logo; How it works, Science, Data and privacy; Log in; primary button "Project my marathon"	Sticky, minimal
Hero: the Split Board	Three inputs (21.1 km split, age, category) drive a live projected finish with the typical-error band from the model card, an evidence badge and a "How is this	[V] Real model behind a public, rate-limited, stateless endpoint. The digits settle once on first load: the page's single orchestrated motion

Section	Content	Notes
	calculated?" link	
The problem	Generic plan versus your plan, two short columns	No statistics beyond the proposal's cited research
How it works	(1) Bring your training, (2) See your projection and what similar runners did next, (3) Fuel the race inside your gut's limit	Numbered because it is a genuine sequence
Fuel gap illustration	Glycogen budget falling, intake lifting it, the 90 g/h ceiling as a line	[P] A concept illustration with example values, labelled as such until the engine is validated
Runners like you	Three case cards	[P] Labelled "illustrative" until the case base is live
Trust band	The published research the system builds on, the open datasets, the evidence-tag legend	Links to /science and /data
For athletes / For coaches	Two short columns, one visual each	Coach column shows the review queue
Privacy promise	Consent, coach sharing you control, export and delete, GPS privacy zones	Plain language
FAQ	What data do I need? Does it cover 5K? (Projection is marathon-only; fueling covers 5K to marathon.) Is this medical advice? How accurate is it?	"How accurate" links to the model card
Final call to action and footer	One button	—

Copy guardrails: accuracy statements come only from model cards; no "AI-powered" hype; sentence case; buttons name the action ("Project my marathon", "Build my fuel plan"). **Engineering:** static rendering, self-hosted fonts, lazily hydrated demo widget, performance budget on Core Web Vitals, privacy-friendly analytics.

3.4 Athlete app pages

Today — one-glance status and the next action.

- Projection tile: latest projected finish, range, evidence badge; sparkline once two or more projections exist.
- This-week ledger: completed volume, long run.
- Next-race Race Ribbon summary.

- Runners-like-you strip (when CBR is live) and a body strip (resting HR, sleep against baseline).
- Attention list: data gaps, pending coach reviews, inputs outside the model's range.
- Empty state: "Add a marathon split to see your projection", plus "Explore with a sample athlete" (clearly labelled synthetic).

Activities — feed and calendar heatmap; import wizard with a validation report (skipped rows, duplicates, timezone issues). **Activity detail:** map with privacy zones, elevation profile, synchronized pace/HR/GAP chart with brushing, splits table, TRIMP, tags, edit and delete, and "used in" (which weekly aggregates include this run).

Train — Projection panel **[V]** (pick a logged marathon or type a split), sensitivity slider, history and calibration. Runners like you **[P]**. What-if explorer **[P]** with the $\pm 5\%$ goal delta and the 10% cap always visible. Every adaptation carries its claim level (S9.6).

Fuel (Fuel Lab) — tabs for pre-race, in-race and post-race. Daily target table by days-to-race; in-race planner with an intake slider that cannot pass 90 g/h (with a "why 90?" explainer on SGLT1 and GLUT5); glycogen trace; inputs panel where advanced values (lower-body fat-free mass, RER, running economy) show **Estimated** unless the athlete supplies them; constant drawer listing every constant with its source and status; scenario compare; export.

Plans — list of drafts and published plans. Plan detail centres on the **Race Ribbon** (S4), then projection, adaptations, nutrition phases, timeline, coach comments, versions and export.

Body — resting HR, sleep and calories with a personal baseline band (rolling mean $\pm$ SD, descriptive). HRV appears only if the athlete's source provides it. There is deliberately **no readiness score**: the proposal defines no such model.

Results — log a race (finish time, optional 21.1 km split); predicted-against-actual chart; error summary next to the model card's validation error; opt-in to contribute the outcome to monitoring.

Reports — weekly and monthly summaries with PDF export.

Settings — profile; units (metric or imperial, pace per km or per mile); privacy (zones, sharing, consent ledger); connections; notifications; data export and deletion; accessibility (reduced motion, text size).

3.5 Coach workspace

- **Roster:** athletes with status chips (data freshness, plan status, pending reviews); sort, filter, bulk invite.
- **Athlete overview:** the athlete's views through a coach lens, plus private notes and "compare with last block".
- **Review queue:** a diff of proposed against current plan; approve, edit or reject with a reason. When an athlete has a linked coach, nothing reaches the athlete until it is approved.
- **Groups:** aggregate volume and adherence; comparisons between named athletes are off by default.
- **Reports:** team summaries and per-athlete PDFs.

3.6 Admin console (internal)

Models (bundles, status, metrics, promote and roll back) · Datasets (registry with licence, provenance, allowed uses) · Users (support-safe views, no raw health data by default) · Flags (evidence-tier rollouts) · Audit (searchable) · links to monitoring dashboards.